JACC
VOL. 87, NO. 25, 2026
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PUBLISHED BY ELSEVIER

EDITORIAL COMMENT

Epicardial Fat Radiomics as a New Frontier in Heart Failure Risk Stratification



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Preventing heart failure (HF) is a problem of timing. Years, sometimes decades, of subclinical biological remodeling precede the hemodynamic and electrical abnormalities that we rely on for detection. Clinical risk scores quantify upstream susceptibility but cannot image the myocardium, circulating biomarkers require thresholds to be crossed before triggering concern, and electrocardiographic surrogates remain insensitive to early structural changes. Even in combination, these tools frequently miss individuals with nascent myocardial damage. Yet, it is precisely this population in whom early intervention would be most impactful.

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One underexplored strategy is to characterize the heart indirectly through its surrounding tissues. Over the past decade, mechanistic work has established that epicardial adipose tissue (EAT) is not merely a passive depot adjacent to the myocardium, but rather an active participant in bidirectional paracrine signaling with the underlying muscle.[1] Inflammatory signals from injured myocardium induce compositional changes in EAT, including lipolysis, fibrosis, and neovascularization.[2] These changes produce compositional shifts detectable on imaging that scale with disease severity. Whether this biology can be captured at scale through routine clinical imaging and translated into a prognostic tool for HF has remained unexplored.

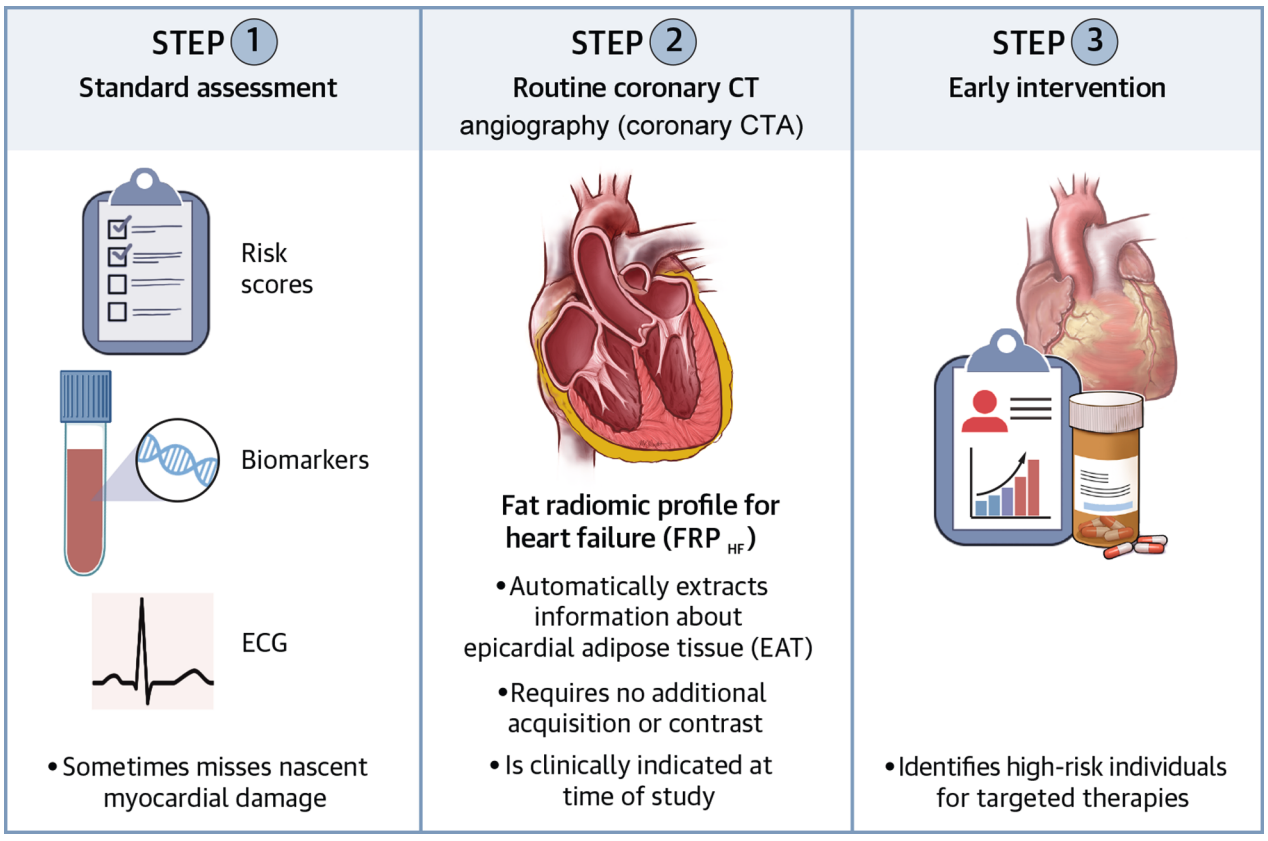
In this issue of *JACC*, Oikonomou et al[3] leveraged this biological framework to develop a fully automated, deep learning-derived radiomic signature referred to as the fat radiomic profile for heart failure (FRP_{HF}). FRP_{HF} captures the texture, geometry, and attenuation heterogeneity of EAT from standard coronary computed tomography angiography (CTA) using 1,655 radiomic features. In a multicenter cohort of 72,751 patients from 9 UK hospitals, the largest coronary CTA radiomic study conducted to date, the authors trained and externally validated FRP_{HF} with a C-statistic of 0.85. Each quartile increase in FRP_{HF} conferred a 4-fold higher adjusted risk of developing HF, and individuals in the highest decile carried a 20-fold excess risk compared with the lowest. Taken together, these findings establish a new dimension to coronary CTA interpretation that moves beyond coronary anatomy to capture biologically grounded signals of future myocardial disease.

The strategic value of anchoring this methodology to coronary CTA is as important as the biological premise. For many practitioners, coronary CTA has become a first-line test for coronary artery disease and now generates large volumes of high-resolution cardiac imaging for indications entirely unrelated to HF. The FRP_{HF} algorithm requires no additional acquisition, no additional contrast, and no reader involvement beyond the index coronary CTA. It converts a clinically obtained scan into a simultaneous HF risk screen at no incremental cost. This paradigm of opportunistic phenotyping during a clinically indicated study parallels recent works leveraging artificial intelligence (AI)-enabled electrocardiography and echocardiography models to detect structural heart disease and stratify long-term

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

ISSN 0735-1097/$36.00
https://doi.org/10.1016/j.jacc.2026.03.160

FIGURE 1 Standard Clinical Tools Frequently Miss Nascent Myocardial Damage



The fat radiomic profile for heart failure (FRP_{HF}) algorithm leverages clinically indicated coronary computed tomography angiography (CTA) to automatically extract epicardial adipose tissue (EAT) radiomics without additional imaging or contrast. This allows for the early identification of high-risk individuals and potential targeted therapeutic interventions. ECG = electrocardiography.

risk from routinely acquired signals and is broadly applicable wherever suitable computational tools can be layered onto existing clinical workflows (Figure 1).[4,5] The fully automated pipeline, operator independence, and demonstrated function across heterogeneous scanner platforms all favor implementation at scale over more resource-intensive phenotyping methods.

> *"Their automated radiomic pipeline transforms a standard coronary CTA examination into a biologically grounded HF risk screen."*

Equally important is what FRP_{HF} captures that conventional risk assessment does not. Body mass index and EAT volume explained only 1.4% and 6.5% of the variance in FRP_{HF}, respectively. The PREVENT-HF (Predicting Risk of CVD EVENTs-Heart Failure) equation accounted for 18% of the variance, leaving the majority of the radiomic signal unexplained by traditional measures. The predictive value of FRP_{HF} also persisted in individuals without diabetes and those with normal body mass index. This implies that it captures a layer of biological risk present below conventional clinical thresholds. These observations have direct implications for opportunistic screening and therapeutic targeting.[6] Previous studies have demonstrated that AI-based screening strategies can efficiently identify primary prevention populations at high risk for targeted therapy when clinical risk scores are combined with biomarkers such as N-terminal pro-B-type natriuretic peptide.[7,8] An imaging signature that identifies elevated HF risk beyond clinical criteria for diabetes or obesity could complement these strategies and expand their reach to individuals who may benefit from more aggressive lifestyle intervention.[9]

One of the more compelling findings is the specificity of FRP_{HF} for myocardial rather than vascular pathobiology. The authors directly compared their global EAT radiomic signature with the pericoronary fat radiomic profile (fat radiomic profile for perivascular adipose tissue), a regional signature they previously developed for coronary inflammation and myocardial infarction risk.[10] The 2 profiles share less than one-quarter of their variance and diverge sharply in their outcome specificity. Notably, FRP_{HF} predicts HF across the full range of ejection fractions and regardless of ischemic heart disease status, whereas the fat radiomic profile for perivascular adipose tissue tracks myocardial infarction and ischemic HF but not HF in the absence of ischemia. That global EAT and pericoronary fat on the same scan distinguish myocardial from vascular pathobiology reinforces the mechanistic basis of the approach.

Several considerations are noteworthy for immediate clinical translation. The derivation and validation cohorts were drawn exclusively from UK National Health Service centers, in which demographics and utilization patterns differ from U.S. populations. Whether FRP_{HF} generalizes to more racially and ethnically diverse populations remains untested and should be a high priority. The incremental gain in discrimination over a baseline model incorporating clinical risk factors and coronary anatomy, while statistically significant and accompanied by clinically relevant net reclassification improvement, was modest in absolute terms. Ultimately, risk stratification is only as valuable as the interventions that it informs. Whether FRP_{HF} changes in response to cardiometabolic therapies, and whether such changes predict outcomes, would help distinguish it from a static risk marker and establish EAT remodeling as a modifiable biological target. Finally, as AI-derived risk signals proliferate across computed tomography, echocardiography, and electrocardiography, how these complementary modalities are integrated, delivered to providers within existing workflows, and translated into actionable treatment decisions will require prospective implementation studies.

In summary, Oikonomou et al[3] should be commended for demonstrating that EAT is a source of prognostic information inaccessible to conventional risk scores or volumetric fat quantification alone. Their automated radiomic pipeline transforms a standard coronary CTA examination into a biologically grounded HF risk screen. Prospective evidence linking FRP_{HF}-guided stratification to improved outcomes is now needed, but the foundation presented in this study makes a compelling case for much closer attention to the epicardial fat depot.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Segar has received honoraria from Idorsia and Otsuka; has received nonfinancial support from Merck; has served as an advisor for AccurKardia; has served on the advisory board for descendantsDNA; and is the founder of ReCODE Medical. Dr Pandey has received research support from the National Institutes of Health, the American Heart Association, Applied Therapeutics, Roche, Ultromics, Gilead Sciences, Bayer, and AstraZeneca; and has received advisor, consultant, or speaker honoraria from Tricog Health, Lilly, Rivus, Roche Diagnostics, Axon Therapies, Edward Lifesciences, Science37, Novo Nordisk, Bayer, Medical AI, AstraZeneca, Baylor Scott and White Research Institute, Boehringer Ingelheim, iRhythm Technologies, Tourmaline Bio, Merck, Sarfez Pharmaceuticals, Ultromics, Kardigan, Tenax Pharma, Alnylam, Abbott, Kieele Health, Anumana, Acorai, Novartis, and Antlia Biosciences.

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KEY WORDS artificial intelligence, cardiac computed tomography, epicardial adipose tissue, heart failure, obesity, radiomics